Secure Hybrid AI for On-Device Anomaly Detection under Trusted Execution: Robust Quantum-Inspired Kernels, Attested Models, and Measurable TEE Trade-offs

Track: Amazon Research Awards — Fall 2026 — AI for Information Security
PI: [Full Name, Title, Affiliation, Email] Co-PI (optional): [Name / N/A]
Cash funding requested: USD $72,000 AWS Promotional Credits requested: USD $28,000
Amazon contact (optional): [Name / N/A]

Abstract

Cyber-physical and edge devices that protect industrial control, logistics, and cloud-adjacent infrastructure increasingly require on-device anomaly detection whose model parameters and inference path remain trustworthy even if the rich execution environment (REE) is compromised. Published quantum-hybrid one-class SVMs report strong HAI ICS results in noiseless simulation, yet when the same fidelity feature map is executed under the paper's stated angle normalization on real ARM Trusted Execution (OP-TEE on Raspberry Pi 5), the kernel collapses and accuracy falls to ~24%. This proposal advances a secure hybrid AI stack that (i) places compact TinyML and calibrated quantum-inspired fidelity kernels inside OP-TEE Trusted Applications, (ii) develops QMOC-SVM—a robustness methodology with bandwidth selection, landmark design, and label-free calibration for one-class / weakly supervised settings, and (iii) adds attested model provisioning, TEE-resident cryptography, and rigorous accuracy–latency–energy–security evaluation across ICS and cloud/device telemetry. Expected outcomes include open-source TA code, public benchmarks, and peer-reviewed results aligned with ARA themes of threat/anomaly detection, learning with limited labels, and AI for device-focused security.

Keywords — trusted execution (OP-TEE); on-device anomaly detection; one-class / weakly supervised learning; quantum-inspired kernels; industrial control security; model confidentiality; edge AI security

1. Introduction and Significance

Research problem. Anomaly detection for cyber-physical systems (CPS) and edge devices must satisfy three constraints at once: (1) severe label scarcity—attacks are rare (~4% in HAI 20.07) and one-class / weakly supervised methods are often mandatory; (2) resource and trust constraints—models and features may leak IP or be tampered with if they reside only in the REE; and (3) reproducible deployability—claims based solely on cloud quantum simulators may not transfer to Secure World memory and latency budgets.

Cultice et al. (arXiv:2409.04935) report 87% accuracy / F1 0.86 on HAI using an 8-qubit fidelity one-class SVM (QOC-SVM) in Qiskit. Our preliminary physical OP-TEE implementation on Raspberry Pi 5 shows that a paper-faithful 3-repetition, $[-\pi, \pi]$ map yields ~24% accuracy (every normal flagged as attack) due to near-orthogonal fidelities, while a calibrated variant (QMOC-SVM: 1 repetition, training-pair bandwidth, more landmarks, calibration-chosen ν) reaches 66.2% accuracy / F1 0.583 as the strongest one-class TEE result—still behind a supervised TinyML baseline at 84% on the same draw. This gap is a scientific opportunity: secure on-device AI needs methods that are robust under TEE constraints, not only accurate in unconstrained simulation.

Significance for Amazon / ARA. The work maps to Fall 2026 AI for Information Security—especially threat/intrusion/anomaly detection, learning with limited or noisy labels, and AI for malware/device security—and complements Amazon Security interests in ARM platform security and on-device monitoring. Publication and open-source TA release match ARA gift norms.

2. Proposed Approach and Novelty

We propose a 12-month program on a Hybrid QAI-in-TEE architecture already prototyped in our `pi5-optee` platform (real OP-TEE on Pi 5; 114/114 `xtest`; dedicated `qai_ta`): REE performs feature extraction; TEE hosts TinyML and fidelity-kernel inference; the host fits classical one-class

SVM on TEE-produced kernels when needed.

2.1 Technical work packages

- WP1 — Robust one-class / weakly supervised detectors for Secure World. Generalize QMOC-SVM: automatic bandwidth and ν selection without test labels; support-vector pruning; approximate / Nyström landmark schemes to cut O(|test|·|train|) TEE kernel cost; compare against RBF OC-SVM, isolation forests, and supervised TinyML under identical TEE budgets.
- WP2 — Trustworthy model lifecycle in OP-TEE. Replace prototype protected paths with TEE AES and secure storage; attested provisioning of model weights/landmarks; integrity measurement of TA binaries; threat model covering REE malware that tampers with features, models, or return paths.
- WP3 — Multi-domain evaluation and AWS-scale telemetry. Extend beyond HAI 20.07 to additional ICS corpora and curated cloud/device log subsets via AWS. Measure detection quality, TEE latency, energy, and memory.
- WP4 — Open artifacts and community benchmarks. Release signed TA images, host clients, preprocessing scripts, fixed seeded evaluation draws, and a public protocol for REE vs TEE parity.

2.2 Novelty

- TEE-native quantum-inspired AD: fidelity kernels as deterministic Secure World statevector code—not a full quantum SDK in a TA—protecting landmarks/support vectors.
- Robustness methodology (QMOC): principled bandwidth/calibration that prevents fidelity collapse under published angle schedules in limited-label regimes.
- Measurement science for secure edge AI: joint reporting of detection quality and TEE cost envelopes that simulators omit, closing the claim–deployment gap.
- Attack-aware evaluation: confidentiality/integrity experiments under an explicit REE-compromise threat model, beyond accuracy-only tables.

3. Evaluation Methodology

- Datasets: HAI 20.07 (primary; MA60, Gini features); ≥1 additional ICS corpus; AWS-hosted synthetic and/or sanitized telemetry for cloud/device anomaly tasks.
- Protocols: fixed seeded draws; natural base-rate evaluation in addition to balanced mixes; calibration sets disjoint from landmarks and test.
- Baselines: classical RBF OC-SVM; supervised TinyML in TA; paper-like QOC-SVM; REE-only vs TEE inference parity.
- Metrics: accuracy, precision, recall, F1, FAR@target; TEE ms/pair and row latency; power/energy; TA image and scratch memory; security analysis vs REE tampering.
- Hardware: Raspberry Pi 5 + real OP-TEE (existing); optional AWS EC2 Graviton for REE-scale ablations—not as a substitute for Secure World claims.

4. Expected Results, Milestones, and Timeline

- Months 0–3: harden QMOC/TinyML TA (AES + secure storage); finalize threat model; extend HAI tables with natural base-rate metrics; technical report v1.
- Months 3–6: landmark/Nyström acceleration; multi-dataset study; AWS pipeline for preprocessing and Gram ablations; draft conference paper.
- Months 6–9: attested provisioning prototype; REE-adversary experiments; open-source code + benchmark release; workshop/demo on Pi 5.
- Months 9–12: journal/extended paper; public splits and leaderboard; ARA status report; optional cloud ↔ edge monitoring notes.

Deliverables: (1) open-source OP-TEE QAI TA + host tooling; (2) calibrated one-class methodology and reference results; (3) peer-reviewed publications; (4) reproducible benchmarks with latency/energy/security dimensions; (5) educational materials for secure edge ML.

5. Funds Needed (USD; no administrative overhead)

Cash — $72,000: graduate research assistant support (~$48,000 stipend/tuition portion directly attributable to this project); PI summer effort (~$12,000); edge hardware, SD imaging, probes/meters for energy measurement (~$5,000); domestic conference travel (~$5,000); open-source CI and artifact hosting (~$2,000).

AWS Promotional Credits — $28,000. Planned products: Amazon EC2 (Graviton/x86) for dataset preprocessing and classical ablations; Amazon S3 for datasets and artifact release; AWS Batch / SageMaker for hyperparameter sweeps and weakly supervised baselines; Amazon CloudWatch for telemetry experiments; modest AWS IoT usage for optional device-stream pilots. Credits amplify scale without replacing on-device TEE evaluation on physical Pi 5 Secure World.

Additional information. This proposal contains no confidential information. Prior Amazon funding: see Appendix C (default: none). PI eligibility and ARA Program Rules will be confirmed at submission. Preliminary platform and HAI/TEE measurements are documented in the open `pi5-optee` repository.

Appendix A — References (does not count toward page limit)

[1] T. Cultice, M. S. H. Onim, A. Giani, and H. Thapliyal, “Anomaly Detection for Real-World Cyber-Physical Security using Quantum Hybrid Support Vector Machines,” arXiv:2409.04935, 2024.

[2] Linaro / OP-TEE, “Open Portable Trusted Execution Environment,” documentation and reference implementations.

[3] ICSDataset, “HIL-based Augmented ICS Security Dataset (HAI 20.07),” https://github.com/icsdataset/hai.

[4] V. Havlíček et al., “Supervised learning with quantum-enhanced feature spaces,” Nature, 2019.

[5] B. Schölkopf et al., “Estimating the Support of a High-Dimensional Distribution,” Neural Computation, 2001 (One-Class SVM).

[6] pi5-optee project documentation: Hybrid QAI in OP-TEE architecture, running manual, and HAI TEE measurement reports (QOC-SVM / QMOC-SVM / TinyML).

[7] Amazon Research Awards, “AI for Information Security call for proposals — Fall 2026,” Amazon Science.

[8] Related ICS anomaly-detection and classical SVM baselines on SWaT/HAI (cf. related work in [1], §II-D).

Appendix B — CV of PI (placeholder; does not count toward page limit)

[PI Full Name] — [Title], [Department], [Institution]. Research interests: trusted execution, edge/embedded security, machine learning for anomaly detection, cyber-physical systems. Replace this appendix with a one-page CV listing appointments and the 5–6 papers most relevant to this proposal.

Most relevant publications (examples to replace):

- [Paper 1 — replace with citation]
- [Paper 2 — replace with citation]
- [Paper 3 — replace with citation]
- [Paper 4 — replace with citation]
- [Paper 5 — replace with citation]
- [Paper 6 — replace with citation]

Co-PI CV (optional): attach one page if a Co-PI is named.

Appendix C — Previously Funded Project Summary

If the PI or project team received cash funding or AWS Promotional Credits directly or indirectly from Amazon within the past 3 years, complete ARA Appendix C (project title; amount and time; source; recipient; remaining credits; summary of results/papers/open-source). Default declaration for this draft: No Amazon cash funding or AWS Promotional Credits in the past three years. Update before submission if applicable.